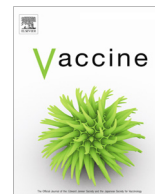




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Replication of live attenuated influenza vaccine viruses in human nasal epithelial cells is associated with H1N1 vaccine effectiveness

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ABSTRACT

In the 2013–2014 and 2015–2016 influenza seasons, live attenuated influenza vaccine (LAIV) generated reduced vaccine effectiveness (VE) against circulating H1N1 strains. This reduced VE coincided with the introduction of pandemic 2009 H1N1 (A/H1N1pdm09) vaccine virus reassortants, in place of pre-2009 seasonal H1N1 strains. Here, we explored one specific hypothesis for reduced VE; decreased replicative fitness of A/H1N1pdm09 strains in humans. Two A/H1N1pdm09 strains with reduced VE, A/California/07/2009 (A/CA09) and A/Bolivia/559/2013 (A/BOL13), were compared to pre-2009 seasonal H1N1 strains, A/New Caledonia/20/1999 (A/NC99) and A/South Dakota/6/2007 (A/SD07). Initial results showed that A/H1N1pdm09 strains had reduced multi-cycle infectivity in Madin-Darby Canine Kidney (MDCK) cells, compared to their pre-2009 counterparts. The A/BOL13 viral titre was found to be 2.65 log₁₀/mL lower when measured by multi-cycle 50% tissue culture infectious dose (TCID₅₀) assay compared to single-cycle fluorescent focus assay (FFA). By contrast, clinically effective A/NC99 titres differed by only 0.54 log₁₀/mL. In human alveolar (A549) cells, A/H1N1pdm09 strains replicated less than pre-2009 strains, with A/CA09 and A/BOL13 generating lower peak viral titres over 5 days. This phenotype was corroborated in physiologically relevant, primary human nasal epithelial cells (hNECs). Here, peak titres for pre-2009 strains A/NC99 and A/SD07 were 8.43 log₁₀ TCID₅₀/mL and 8.52 log₁₀ TCID₅₀/mL, respectively, versus 6.89 log₁₀ TCID₅₀/mL and 6.06 log₁₀ TCID₅₀/mL for A/H1N1pdm09 strains A/CA09 and A/BOL13. This confirmed a reduced ability of A/H1N1pdm09 strains to sustain replication in human respiratory cells. Using this information, H1N1 candidate A/Slovenia/2903/2015 (A/SLOV15) was characterised for replacement of A/BOL13 in the 2017/18 LAIV. A/SLOV15 produced comparable single and multi-cycle infectivity titres (Δ 0.16 log₁₀/mL) and reached a peak titre 1.23 log₁₀ TCID₅₀/mL higher than that of A/BOL13 in hNEC cultures. Taken together, these data suggest a reduction in sustained multi-cycle replication in human cells as a plausible root cause for reduced A/H1N1pdm09 VE.

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1. Introduction

On an annual basis, seasonal influenza can cause approximately 3–5 million cases of severe illness and around 290,000–650,000 deaths worldwide [1]. Live attenuated influenza vaccine (LAIV) provides annual immunisation against seasonal circulating influenza strains, helping prevent this illness and mortality. Unlike inactivated vaccines, LAIV is administered intranasally and elicits a mucosal as well as humoral immune response [2]. LAIV viruses are produced in embryonated chicken eggs [3] and

formulated as a quadrivalent vaccine (QLAIV) containing one component from each of the influenza A H1N1 and H3N2 subtypes and both of the influenza B lineages, B Yamagata and B Victoria. The master donor virus (MDV) backbones used to generate LAIV strains are cold adapted, temperature sensitive and attenuated. Upon vaccination, these phenotypes allow replication of LAIV viruses in the colder temperatures of the human nasal passage, while restricting replication in the warmer temperatures of the lower respiratory tract [4]. The capacity of LAIV to replicate well in the upper respiratory tract is essential for the generation of the humoral and cellular immune responses, both local and systemic, which confer protection from subsequent influenza virus infection [5,6].

Meta-analyses of real-world vaccine effectiveness (VE) data obtained between 2009 and 2016 showed that LAIV was consistently effective against influenza B viruses. LAIV was also effective

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against antigenically matched influenza A H3N2 strains, with VE estimates comparable to inactivated influenza vaccines (IIV) [7,8].

However, following the 2013–2014 influenza season, it was recognised that the A/H1N1 pandemic 2009 (A/H1N1pdm09) component of QLAIV, A/California/07/2009 (A/CA09), had provided reduced VE, relative to IIV, in the US, with a consolidated VE estimate of 19% in 2013–2014 [7]. Initial findings suggested that reduced A/CA09 VE may have been due to poor thermostability of the A/CA09 haemagglutinin (HA) protein [9]. Increased outdoor temperature during LAIV unloading at US distribution centres was found to correlate statistically with reduced VE of A/H1N1pdm09 viruses [10]. As a result, A/Bolivia/559/2013 (A/BOL13), a strain with improved thermostability, was included in the 2015–2016 LAIV vaccine [11]. Despite improved thermostability, A/BOL13 still provided reduced VE when compared with IIV in the same year, with consolidated VE estimates of 32% for LAIV versus 72% for IIV [7]. Consequently, the Advisory Committee on Immunization Practices voted not to recommend FluMist Quadrivalent® for use in the US in the 2016–2017 and 2017–2018 influenza seasons [12,13].

Recently, we showed that HA protein stability was unlikely to be the sole determinant of H1N1 VE, as the HA thermal and pH stability profiles of the clinically suboptimal A/BOL13 were comparable with previously effective influenza A LAIV strains, including A/New Caledonia/20/1999 (A/NC99) [14]. This suggested that there might be another contributing factor common to A/H1N1pdm09 strains leading to their reduced VE.

To understand this contributor in more detail and determine the root cause, several different hypotheses were explored. The LAIV manufacturing and distribution process was examined in detail. Scrutiny was first applied to the vaccine supply chain, including shipment, handling and storage. However, no root cause could be identified, and a thorough scientific investigation of vaccine virus characteristics was initiated. Neither changes in the biophysical properties (such as morphology and size) of A/H1N1pdm09 viruses nor a potential antigenic mismatch to circulating strains were found to be involved (unpublished data, [15]). Pre-existing anti-influenza immunity was also ruled out, as children vaccinated in previous years may have an improved humoral immune response compared with children vaccinated for the first time [16,17]. Thus, two primary hypotheses to explain reduced A/H1N1pdm09 VE remained: vaccine–virus interference in the QLAIV formulation and a decrease in LAIV replicative fitness in human cells [18]. In this study, we aimed to determine if a decrease in replicative fitness of LAIV A/H1N1pdm09 strains could provide an explanation for their reduced VE.

To address this hypothesis, pre-2009 LAIV strains A/NC99 (with proven clinical efficacy) [19,20] and A/South Dakota/6/2007 (A/SD07, a component of the 2008–2009 commercial vaccine formulation), were compared with A/H1N1pdm09 strains, A/CA09 and A/BOL13, with reduced VE [7,21]. Here, we demonstrated that both A/CA09 and A/BOL13 possess deficiencies in multi-cycle infectivity in Madin–Darby Canine Kidney (MDCK) cells, relative to their pre-2009 counterparts. We also showed that both A/H1N1pdm09 strains replicate to significantly lower levels in human alveolar A549 cells and, most importantly, in physiologically relevant human nasal epithelial cell (hNEC) cultures. Finally, to ensure use of the most effective strains possible, we assessed the replicative fitness of a novel A/H1N1pdm09 vaccine candidate, A/Slovenia/2903/2015 (A/SLOV15), to determine its suitability as a replacement for A/BOL13 in the 2017–2018 QLAIV.

These data indicate that failure to achieve sustained multi-cycle replication in human cells provides a plausible root cause for the reduced A/H1N1pdm09 VE observed. In addition, we suggest that primary hNEC cultures can model H1N1 LAIV virus fitness in vaccine recipients and be a predictive tool to aid the selection of seasonal vaccine strains with increased effectiveness.

2. Materials and methods

2.1. Cell culture

MDCK cells were purchased from the European Collection of Authenticated Cell Cultures and were passaged in Eagle's Minimum Essential Media (EMEM) containing non-essential amino acids (NEAA) and sodium pyruvate (Lonza, Basel, Switzerland) supplemented with 2 mM L-glutamine (Life Technologies, Carlsbad, CA, USA), 10% heat-inactivated foetal bovine serum (FBS) (VWR, Radnor, PA, USA) and 50 µg/mL gentamycin (Life Technologies). A549 cells were purchased from the American Type Culture Collection and were passaged in Dulbecco's Modified Eagle's Media (Life Technologies) supplemented with 2 mM L-glutamine (Life Technologies), 10% heat-inactivated FBS (VWR) and 10 × penicillin–streptomycin (Lonza). Both cell lines were plated into non-coated 24- or 96-well tissue culture plates and were incubated at 37 °C and 5% CO₂ until infection.

Primary hNEC cultures (Epithelix, Geneva, Switzerland) were purchased as fully differentiated, airway epithelial cells and were grown on an air–liquid interface. hNEC cultures closely mimic the physiology and functionality of the human nasal tract [22]. This includes functioning cilia, mucus production, active ion transport and cytokine/chemokine release [23]. hNEC cultures were sourced from pools of at least five healthy human donors to avoid donor-specific effects. Generation of hNEC cultures has been described previously [24]. hNECs were maintained on the basal layer with MucilAir™ culture media (Epithelix), which was changed every 2–3 days. The apical layer was left exposed to air with apical washes performed every 7–10 days to remove mucus. Prior to LAIV infection, hNEC inserts were placed in 24-well tissue culture plates and incubated at 37 °C and 5% CO₂ for up to 2 days.

2.2. Viruses

LAIV viruses were generated using a reverse genetics method, as previously described [25], using MDV A/AnnArbor/6/60 for influenza A strains and B/AnnArbor/1/66 for influenza B strains to provide PB2, PB1, PA, NP, M and NS genes and a strain-specific wild-type (wt) virus for the HA and neuraminidase (NA) genes. During cloning of wt HA and NA genes, primers annealing only to conserved, subtype-specific, sequences were used. Consequently, 5' and 3' untranslated regions for each strain were identical to wt virus. The LAIV viruses included in this study were: A/NC99, A/SD07, A/CA09, A/BOL13, A/Slovenia/2903/2015 (A/SLOV15), A/Panama/2007/1999 (A/PAN99), B/Massachusetts/2/2012 (B/MASS12), B/Phuket/3073/2013 (B/PHU13), and B/Brisbane/60/2008 (B/BRIS08).

Reassortant viruses were inoculated into the allantoic cavity of specific pathogen-free embryonated chicken eggs (Charles River Laboratories, Wilmington, MA, USA) at a titre of 3.1 log₁₀ fluorescent focus units/mL (FFU/mL). Allantoic fluid was harvested after 48–72 h (depending on the optimal incubation period for the strain), pooled and stabilised with 10 × sucrose phosphate (10 × SP, 200 mM sucrose, 3.8 mM potassium di-hydro phosphate and 7.2 mM di-potassium hydrogen phosphate, pH 7.2) to a final concentration of 1 × SP before storage at –80 °C. The viral titres of all LAIV strains were determined by fluorescent focus assay (FFA) in MDCK cells.

2.3. FFA assay

FFA viral growth medium (FVGM) was prepared from EMEM containing 1 × NEAA (Biowest, Nuaille, France) supplemented with 1.1 g/L sodium pyruvate (Lonza), 2 mM L-glutamine (Life Technologies), 50 µg/mL gentamicin sulphate (Life Technologies) and

0.5 µg/mL amphotericin B antimycotic (Life Technologies). Confluent MDCK cells in 96-well tissue culture plates were washed twice with FVGM and duplicate wells were inoculated with 100 µL of a 1:10 serial dilution of virus stock in FVGM. The cells were incubated at 33 °C and 5% CO₂ for 20 h, then washed with FVGM and fixed with 80% acetone in water for 1 h at –20 °C. The cells were then dried at 36 °C for 30 min and washed with phosphate buffered saline (PBS) containing 0.5% Tween (PBST, Sigma-Aldrich, St Louis, Missouri, USA) before labelling for 1 h at 36 °C with a sheep primary antibody against the specific HA protein of the virus (obtained from the National Institute for Biological Standards and Control, UK or the Therapeutic Goods Administration) in 1% bovine serum albumin (BSA) in PBS. Cells were then washed three times with PBST and incubated with Alexa Fluor 488 donkey anti-sheep IgG (Life Technologies, Catalogue # A-11015) in 1% bovine serum albumin in PBS for 1 h at 36 °C, then washed twice with PBST and dried at room temperature for 30 min. An Eclipse Ti inverted fluorescence microscope (Nikon, Tokyo, Japan) was used to determine the mean number of HA-positive foci in two replicate wells at a suitable dilution for counting and the concentration of FFU in the original virus stock was then calculated.

2.4. Virus infection

For MDCK cells and A549 cells in 24-well plates, virus inocula were prepared to an approximate multiplicity of infection (MOI) of 0.0001 and 0.1, respectively. Virus inoculum was prepared with cell-specific growth media (see above), without FBS but with the addition of 1:400 10 × TrypLE (Life Technologies). Cells were inoculated with virus and incubated at 33 °C and 5% CO₂ for 1 h. Post incubation, inocula were removed and cells were washed three times with cell-specific media. The media in each inoculated well was replaced with fresh media and the plates were returned to the incubator at 33 °C and 5% CO₂. Media samples were collected from the cells once every 24 h for 5 days, from the time the cells were placed back in the incubator. Media samples were stabilised with 10 × SP to a final concentration of 1 × SP and stored at –80 °C until 50% tissue culture infectious dose (TCID₅₀) assays were performed.

For hNEC cultures in 24-well plates, virus inoculum was prepared to an approximate MOI of 0.1. Media was placed on the apical surface and the cells were incubated for 20 min at 37 °C and 5% CO₂. Post-incubation, media was removed, and cells were inoculated on the apical surface for 1 h at 33 °C and 5% CO₂. Inocula were removed and the apical surface of the cells was washed three times with cell-specific media (leaving no media present). Media in contact with cells on the basal surface was also changed prior to incubation of cells at 33 °C and 5% CO₂ for 5 days. Apical wash samples were collected from the cells once every 24 h, from the time the cells were placed back in the incubator, for 5 days, by addition of cell-specific media to the apical surface and incubation for 20 min at 33 °C and 5% CO₂ before collection. Samples were stabilised with 10 × SP to a final concentration of 1 × SP and stored at –80 °C until TCID₅₀ assays were performed. To control for any residual hNEC donor variability, A/NC99 was uniformly included as a positive control in all independent infection experiments.

2.5. TCID₅₀ assay

MDCK cells were grown in 96-well tissue culture plates until confluent. Cells were washed twice with cell culture media and inoculated with 1:10 dilutions of virus stock from a pre-prepared serial dilution block containing EMEM media with non-essential amino acids and sodium pyruvate (Lonza) and supplemented with 2 mM L-glutamine (Life Technologies), 50 µg/mL gentamycin (Life Technologies) and 1:400 10 × TrypLE (Life Technologies). Diluted virus (200 µL) was added to wells in four replicates per dilution

and the plates were incubated at 33 °C and 5% CO₂ for 6 days. Cells were then scored for cytopathic effects on an inverted light microscope and titres determined by the Reed-Muench method [26].

2.6. Single-cycle infection immunofluorescence

Inocula were prepared using cell-specific growth media before addition to MDCK and A549 cells at an estimated MOI of 5. Cells were incubated at 33 °C and 5% CO₂ for 16 h in the absence of trypsin, to ensure complete viral entry and protein expression by all strains, irrespective of replication kinetics. Cells were fixed with 4% formaldehyde in PBS at room temperature for 10 min, permeabilised with 1% Triton X-100 (Sigma-Aldrich) in PBS and blocked with 5% FBS in PBS. Cells were then incubated with mouse anti-influenza A nucleoprotein (NP) primary antibody (BioRad, Hercules, CA, USA) followed by goat anti-mouse secondary antibody with Alexa Fluor 488 (Life Technologies, Catalogue # A-11001). All antibodies were diluted in PBS with 5% FBS. Hoechst stain (pentahydrate bis-benzamide, Life Technologies) diluted in PBS was added to cells for 10 min prior to imaging. Inoculated cells were imaged, and the number of fluorescent NP foci quantified using an Image Xpress Micro 4 imaging system with a 4 × objective (Molecular Devices, LLC, San Jose, CA, USA). Cellular DNA and NP staining were visualised using 4',6-diamidino-2-phenylindole and fluorescein-isothiocyanate filters, respectively. For each infection, NP-positive, infected cells were enumerated as a percentage of total 4',6-diamidino-2-phenylindole-positive cells, to normalise for variations in cell numbers. This was performed using MetaXpress analysis software (MetaXpress, Reading, UK; Molecular Devices). Relative infectivity of a virus in A549 versus MDCK cells was then calculated by expressing A549 infectivity as a percentage of MDCK infectivity (e.g., a relative infectivity of 100% would indicate equal infectivity in A549 as MDCK). Six independent experiments were performed, with two replicates per experiment.

2.7. Data and statistical analysis

All graphs were generated in GraphPad Prism, version 7.0 (GraphPad Software Inc, San Diego, CA, USA). Statistical analyses were performed in JMP v13 (JMP Software, SAS Institute Inc, Marlow, Bucks, UK). The peak daily titre values (subsequently referred to as peak titre) of viral growth curves were generated from at least three experiments with three replicates per experiment per strain. FFA and TCID₅₀ values are displayed as virus titre Log₁₀/mL. Peak daily titre values were compared using the Tukey's honest significant difference test after least squares mean regression analysis of nested replicate and repeat results (subsequently referred to as the Tukey analysis), with *p < 0.05 considered significantly different. Comparison of virus titre at individual timepoints for growth curves was conducted by ANOVA for all timepoints. Following identification of significance by ANOVA (*p < 0.05), individual timepoints were compared by Tukey's honest significant difference test, with *p < 0.05 considered significantly different. Immunofluorescence studies were analysed by student's unpaired t-test, with *p < 0.05 considered significantly different. Welch's test with Bonferroni correction was used to determine significant differences between FFA and TCID₅₀ titres with unequal variance. A corrected p-value of *p < 0.006 (p < 0.05/8) was used to account for eight comparisons within a single dataset [27].

3. Results

3.1. A/H1N1pdm09 LAIV strains are deficient in multi-cycle infectivity

Initially, a panel of LAIV strains with published VE data including influenza B subtype viruses (B/MASS12, B/PHU13 and

B/BRIS08), H3N2 virus A/PAN99 and H1N1 viruses (A/NC99, A/SD07, A/CA09 and A/BOL13) were analysed by both FFA and TCID₅₀ assays. FFA quantifies individual cells expressing HA protein 20 h after infection but does not account for release and spread of the virus. TCID₅₀ requires spread of the virus over multiple replication cycles to generate visible cytopathic effects on the MDCK cell monolayer. These methods therefore indicate virus infectivity based on single-cycle or multi-cycle replication, respectively. This comparison revealed a greatly reduced ability of A/CA09 and A/BOL13 to support multiple rounds of replication, with TCID₅₀ titres significantly lower than FFA titres (Fig. 1). The TCID₅₀ titres for A/CA09 and A/BOL13 were 1.19 log₁₀/mL and 2.65 log₁₀/mL lower, respectively, than the FFA titres. Much smaller and non-significant differences were observed for the pre-2009 H1N1 strains, with TCID₅₀ titres for A/NC99 and A/SD07 0.54 log₁₀/mL and 0.60 log₁₀/mL lower than the FFA titres, respectively. The smallest differences observed between FFA and TCID₅₀ titres were for influenza B virus strains, with the greatest difference of these observed for B/MASS12 at 0.30 log₁₀/mL. A larger difference in mean TCID₅₀ titre compared with FFA titre was also observed for A/PAN99 at 1.44 log₁₀/mL, but this was not statistically significant due to the high variance of TCID₅₀ data for this strain. These initial findings highlighted a potential reduction in the ability of A/H1N1pdm09 strains to sustain multiple rounds of replication when compared with pre-2009 H1N1 strains.

3.2. A/H1N1pdm09 strains have delayed replication in MDCK cells

To investigate this further, multi-cycle growth kinetic studies were performed in MDCK cells to determine whether A/H1N1pdm09 strains showed reduced replicative fitness compared with pre-2009 H1N1 strains. Pre-2009 H1N1 strains A/NC99 and A/SD07, and A/H1N1pdm09 strains A/CA09 and A/BOL13, were used

to infect MDCK cells for 5 days, with media samples collected from the cells every 24 h. Following observations that FFA titres might not accurately represent the fully infectious, replication competent virus population of A/H1N1pdm09 strains, infectious virus quantities in cell media were measured by TCID₅₀ assay. These data are shown as growth curves in Fig. 2. All four LAIV strains achieved similar peak viral titres of approximately 8.1–8.5 log₁₀ TCID₅₀/mL (Fig. 2A). However, both A/H1N1pdm09 strains exhibited a delay in accumulation of infectious progeny, peaking on day 4, 1 day later than the pre-2009 H1N1 strains. At day 5 post-infection, A/H1N1pdm09 titres had declined relative to day 4, confirming that peak replication had been achieved. This delay in replication corresponded with the reduced multi-cycle infectivity seen in Fig. 1 and indicated a reduction or delay in the replicative ability of A/H1N1pdm09 strains. A/NC99 titres at days 2 and 3 post-infection were significantly higher than those for both A/CA09 and A/BOL13 (A/CA09: day 2p = 0.01, day 3p = 0.036; A/BOL13: day 2p = 0.011, day 3p = 0.016), confirming a replicative delay of the A/H1N1pdm09 strains relative to this strain. A/SD07, while qualitatively similar to A/NC99, was not statistically distinct from the A/H1N1pdm09 strains. Peak titres for H1N1 strains were not significantly different from each other (Fig. 2B), indicating an equivalent maximum replicative capacity in MDCK cells.

3.3. A/H1N1pdm09 LAIV strains have reduced replicative fitness in A549 cells

Having seen evidence of delayed replication of A/H1N1pdm09 strains in MDCK cells, we aimed to determine whether this was also true in a human respiratory cell line. A549 cells are a human alveolar carcinoma cell line and are thus more representative of the site of LAIV replication *in vivo*. To begin to assess potential differences in replication in A549 relative to MDCK cells, single-cycle infection of LAIV H1N1 strains was compared in both cell types. MDCK cells infected with A/NC99 or A/BOL13, as representative pre-2009 and A/H1N1pdm09 viruses, respectively, exhibited qualitatively similar NP staining frequencies (Fig. 3A, top panels), confirming the MDCK-derived FFA titres shown in Fig. 1. However, there was a visible reduction in infectivity for A/BOL13 in A549 cells when compared with A/NC99 (Fig. 3A, bottom panels). After high-content image analysis, as described in the materials and methods, the infectivity of each virus in A549 cells relative to MDCK cells was calculated. A/NC99 infectivity in A549 cells was similar to that measured in MDCK cells, while A549 infectivity for A/BOL13 was significantly reduced, at only 25% of that in MDCKs (Fig. 3B). This showed that A/NC99 had a significantly higher relative infectivity in A549 cells compared with A/BOL13 (Fig. 3B).

The reduced single-cycle infectivity observed in A549 cells for A/H1N1pdm09 strain A/BOL13 prompted assessment of multi-cycle replication studies for all four H1N1 strains. Pre-2009 H1N1 strains A/NC99 and A/SD07 and A/H1N1pdm09 strains A/CA09 and A/BOL13 were used to infect A549 cells for 5 days, with media samples collected every 24 h. Virus titres in cell media samples were measured by TCID₅₀ assay and growth curves generated. The pre-2009 H1N1 strains A/NC99 and A/SD07 both reached peak viral titres at 4 days post-infection (Fig. 3C), with titres of 7.17 log₁₀ TCID₅₀/mL and 6.62 log₁₀ TCID₅₀/mL, respectively (Fig. 3D). Viral titres obtained for A/H1N1pdm09 strains were still increasing at day 5 post-infection. As such, a true peak titre for A/CA09 and A/BOL13 could not be defined under these infection conditions and would have required longer incubation periods. However, it was clear that the rate of replication of the A/H1N1pdm09 strains was reduced relative to pre-2009 H1N1 viruses, leading to a delay in virus accumulation (Fig. 3C). Over 5 days of infection, maximum titres of 5.79 log₁₀ TCID₅₀/mL and 3.32 log₁₀ TCID₅₀/mL were reached for A/CA09 and A/BOL13, respectively (Fig. 3D).

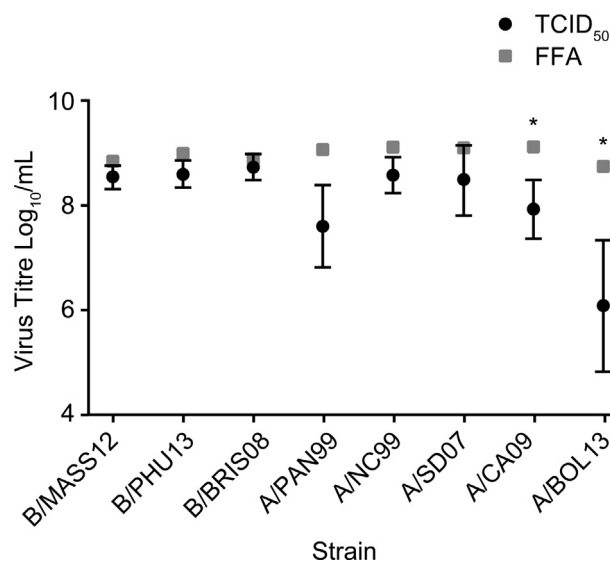


Fig. 1. Fluorescent focus assay (FFA) and 50% tissue culture infectious dose (TCID₅₀) comparison for a panel of live attenuated influenza vaccine (LAIV) strains. Single- and multi-cycle infectivity for a panel of LAIV strains was determined by FFA and TCID₅₀ assay, respectively. For FFA assays, Madin-Darby Canine Kidney (MDCK) cells were infected with LAIV by serial dilution on 96-well plates for 20 h at 33 °C. Infected cells were identified by viral haemagglutinin protein staining and counted. For TCID₅₀ assays, MDCK cells were infected with LAIV for 6 days at 33 °C. Individual wells were scored as cytopathic effects positive or negative and titres determined by the Reed-Muench method [26]. Mean virus titres are shown for both methods from at least six independent experiments. Error bars show ± 1 standard deviation. Differences between mean FFA and TCID₅₀ values for each strain were assessed by Welch's test with Bonferroni correction (due to unequal variance between assays, *p < 0.006 was considered as significant).

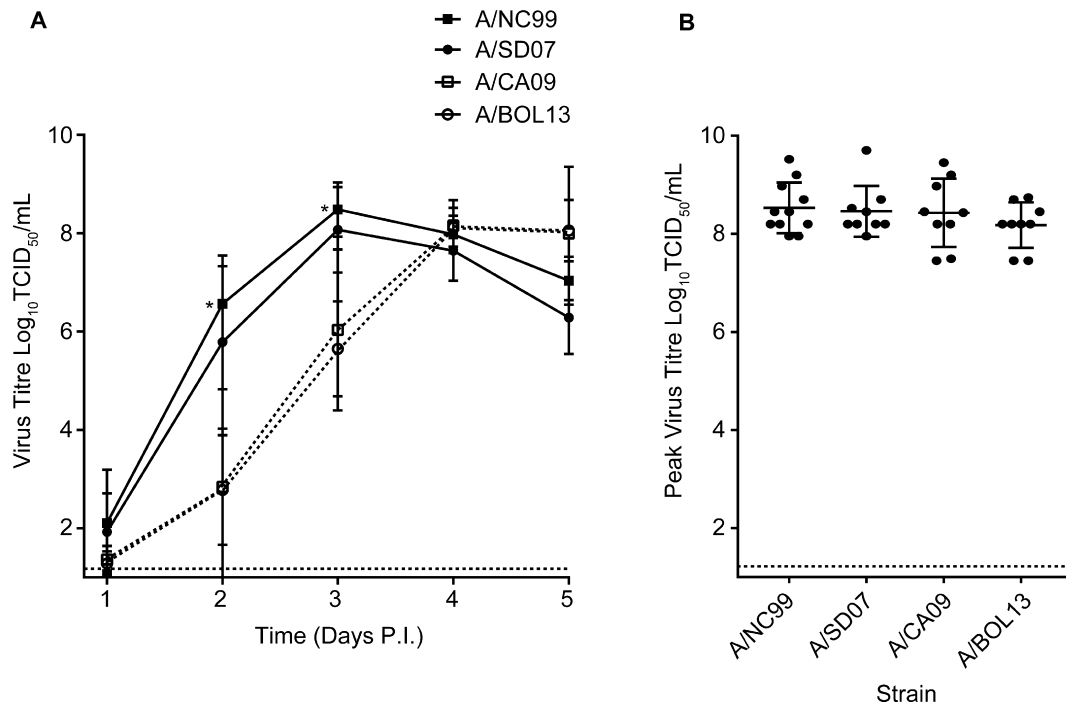


Fig. 2. Virus replication of H1N1 strains in Madin-Darby Canine Kidney (MDCK) cells. MDCK cells were infected with A/NC99, A/SD07, A/CA09 or A/BOL13 at a multiplicity of infection (MOI) of 0.0001 for 5 days at 33 °C with cell media sample collection every 24 h. Viral titres were determined by 50% tissue culture infectious dose (TCID₅₀) assay in MDCK cells. (A) Mean virus titres for each time point were generated from three replicates with at least three independent experiments for each strain and (B) peak viral titres were plotted. Error bars show ± 1 standard deviation and the limit of detection of the assay is shown by the dotted line. Differences between the peak viral titre of each strain were assessed by Tukey analysis (* $p < 0.05$ was considered as significant); *denotes significance compared with the A/CA09 and A/BOL13 strains. P.I., post-infection.

These data demonstrate that A/H1N1pdm09 strains have reduced replicative fitness in human A549 cells compared with their pre-2009 H1N1 counterparts, potentially due to delayed replication kinetics. They also indicate that fitness differences might be host species dependent, with markedly greater distinctions in virus replication apparent in human lung cells rather than canine kidney cells.

3.4. A/H1N1pdm09 LAIV strains have reduced replicative fitness in hNEC cultures

Given the observation that A/H1N1pdm09 strains had reduced replicative fitness in A549 cells, a cell culture system more closely resembling the site of LAIV virus replication in a vaccine recipient was explored. Primary hNEC cultures were infected with pre-2009 H1N1 strains A/NC99 and A/SD07 and A/H1N1pdm09 strains A/CA09 and A/BOL13, as described in the materials and methods. Pre-2009 A/H1N1 strains A/NC99 and A/SD07 resulted in comparable hNEC growth curves (Fig. 4A), with peak titres of 8.43 log₁₀ TCID₅₀/mL and 8.52 log₁₀/mL, respectively, reached at day 3 post-infection (Fig. 4B). Both A/H1N1pdm09 strains demonstrated reduced replication compared with their pre-2009 H1N1 counterparts (Fig. 4A) with significantly lower peak titres of 6.89 log₁₀ TCID₅₀/mL for A/CA09 and 6.06 log₁₀ TCID₅₀/mL for A/BOL13 (~300-fold decrease relative to pre-2009 H1N1 strains). These results indicate that A/H1N1pdm09 LAIV strains have considerably reduced replicative fitness when compared with pre-2009 H1N1 strains in clinically relevant hNEC cultures.

3.5. A/SLOV15 has improved multi-cycle infectivity and replicative fitness in human cells compared with previous A/H1N1pdm09 strains

To determine whether a novel A/H1N1pdm09 vaccine candidate, A/SLOV15, possessed improved replicative fitness relative to its

predecessor, A/BOL13, A/SLOV15 was analysed for growth in hNEC cultures. hNEC cultures were infected with A/SLOV15 at a MOI of 0.1 and incubated for 5 days, as described previously. The results were compared with those for an A/H1N1pdm09 strain with reduced VE, A/BOL13, at one standard deviation above and below the mean for each of the 5 days (Fig. 5A). The growth curves illustrated that A/SLOV15 was able to achieve higher mean viral titres on each day of infection compared with A/BOL13 and that these viral titres were greater than the upper standard deviation for A/BOL13. The peak viral titre obtained for A/SLOV15 in hNEC cultures was 7.29 log₁₀ TCID₅₀/mL, which was significantly higher than that achieved for A/BOL13, at 6.06 log₁₀ TCID₅₀/mL (Fig. 5B). This result confirmed that the new H1N1 vaccine candidate A/SLOV15 possessed improved replicative fitness in hNECs compared with the prior vaccine strain A/BOL13.

The A/SLOV15 viral titre was measured by both FFA and TCID₅₀ in MDCK cells and FFA:TCID₅₀ ratios were calculated (Fig. 5B). The FFA:TCID₅₀ ratio for A/SLOV15 was 0.16, which was considerably lower than the FFA:TCID₅₀ ratio of 2.65 for A/BOL13.

Together, these data demonstrate that the A/H1N1pdm09 strain A/SLOV15 has an improved ability to sustain multi-cycle replication compared with the prior H1N1 vaccine candidate A/BOL13. These findings suggest that A/SLOV15 could generate improved VE outcomes in the clinic.

4. Discussion

Here, we aimed to identify the root cause of the reduced VE of the A/H1N1pdm09 strains A/CA09 and A/BOL13 in the quadrivalent FluMist[®] vaccine. Our initial findings highlighted a deficiency in the ability of the A/H1N1pdm09 strains to initiate multi-cycle infection, with reduced TCID₅₀ infectivity titres compared with single-cycle FFA titres. Subsequent studies in A549 cells suggested a reduced ability of A/H1N1pdm09 strains to infect and replicate in

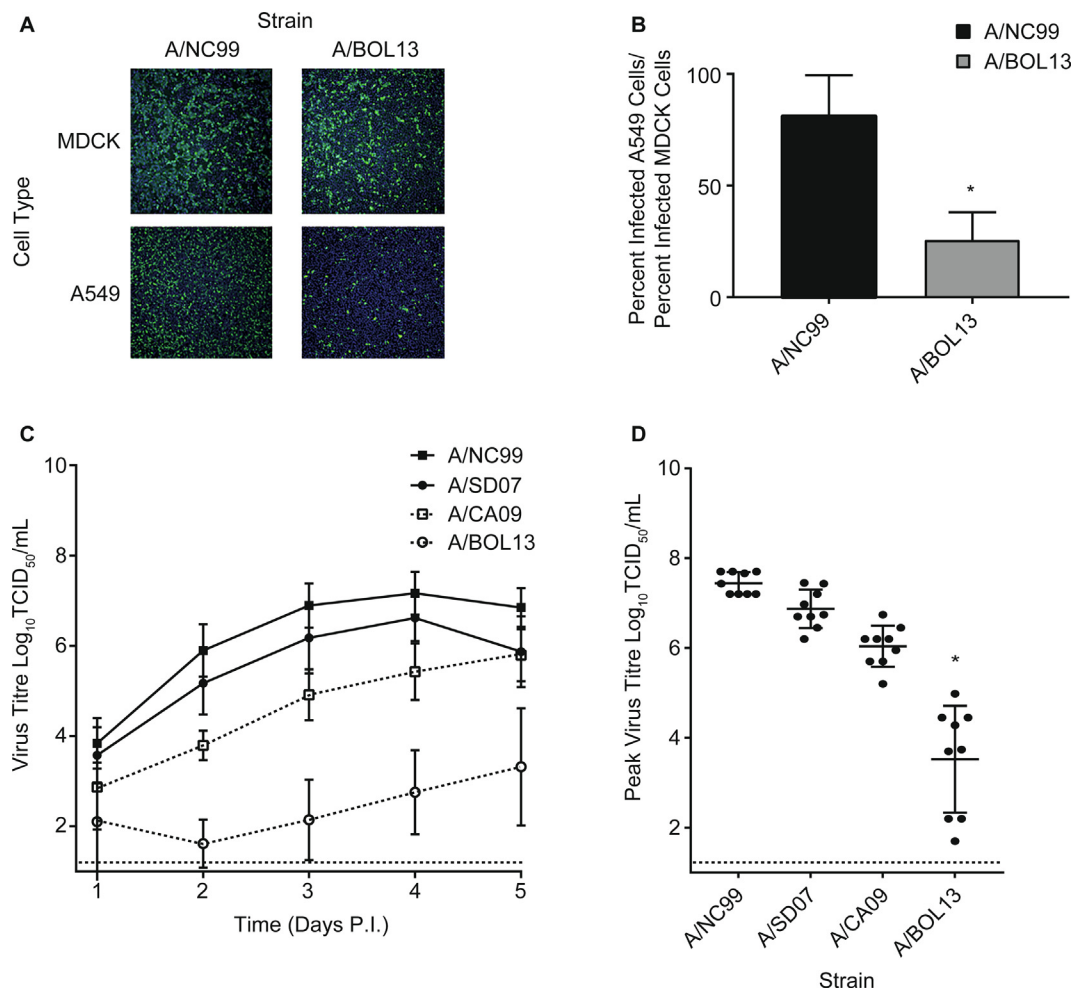


Fig. 3. Virus infectivity and replication of H1N1 strains in A549 cells. (A) Madin-Darby Canine Kidney (MDCK) and A549 cells were infected with A/NC99 and A/BOL13 at a multiplicity of infection (MOI) of 5 for 16 h at 33 °C. All cells were identified by nuclear DNA Hoechst stain (blue) and infected cells by viral nucleoprotein (NP) staining (green) and (B) the percentage of infected A549 cells relative to MDCK cells plotted. Significant differences between strains were determined by Student's *t*-test (unpaired, two tailed, **p* < 0.05). Representative images are presented, but mean values are plotted with error bars depicting ± 1 standard deviation from six independent experiments. (C) A549 cells were infected with A/NC99, A/SD07, A/CA09 or A/BOL13 at a MOI of 0.1 for 5 days at 33 °C with cell media sample collection every 24 h. Viral titres of cell media samples were determined by 50% tissue culture infectious dose (TCID₅₀) assay in MDCK cells. Mean virus titres for each time point were generated from three replicates with at least three independent experiments for each strain and (D) peak viral titres were plotted. Error bars show ± 1 standard deviation and the limit of detection of the assay is shown by the dotted line. Differences between the peak viral titre of each strain were assessed by Tukey analysis (**p* < 0.05 was considered as significant). A/BOL13 had significantly lower peak titres than A/NC99 and A/SD07. P.I., post-infection. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

human cells, relative to clinically effective pre-2009 A/H1N1 strains.

These results were reproduced in hNEC cultures, where A/H1N1pdm09 strains A/CA09 and A/BOL13, with reduced VE, generated lower peak viral titres over a 5-day time course. From these data, we conclude that A/H1N1pdm09 LAIV strains are deficient in generating high levels of sustained virus replication in human nasal airway cells.

Analysis of growth curves in both MDCK and A549 cells also indicated a delay in the time taken for A/H1N1pdm09 strains to achieve peak titre. In both cases, pre-2009 A/H1N1 strains peaked earlier post-infection than their A/H1N1pdm09 equivalents. Ultimately, in MDCK cells, similar peak titres were reached, 1 day later than pre-2009 H1N1 strains. In A549 cells, A/H1N1pdm09 strains failed to reach replicative peak within the 5-day time course used. This suggested longer incubation periods would have been necessary to see true A/H1N1pdm09 peak titres. However, it is possible that irrespective of peak titre, a delay in replication kinetics could prove disadvantageous *in vivo* in the context of a developing immune response. The reduction in single-cycle A549 infectivity

seen for A/BOL13 could help explain the slower replication kinetics in this cell type, as this deficiency would be compounded over multiple replication cycles. While single cycle infectivity was not also measured for A/CA09, it will be important to gain a more detailed understanding of any possible deficiencies in the early stages of the A/H1N1pdm09 LAIV life cycle that might lead to delayed or reduced replication.

Taking these data together, we conclude that reduced replicative fitness of A/CA09 and A/BOL13 provides a plausible root cause for the reduced VE observed for A/CA09 and A/BOL13 LAIV strains [7,21].

FFA is the potency release test for the commercial FluMist® vaccine and remains a key assay in the selection of candidate vaccine strains due to its precision, speed and reliability. However, FFA is a measure of single-cycle infectivity and may therefore preferentially favour viruses with an increased ability to bind, internalise and replicate RNA rather than those which support replication over multiple cycles. Our experiments all therefore utilised TCID₅₀ as a measure of virus replication, to compare synthesis of fully infectious, replication-competent virus, as FFA titres may have masked

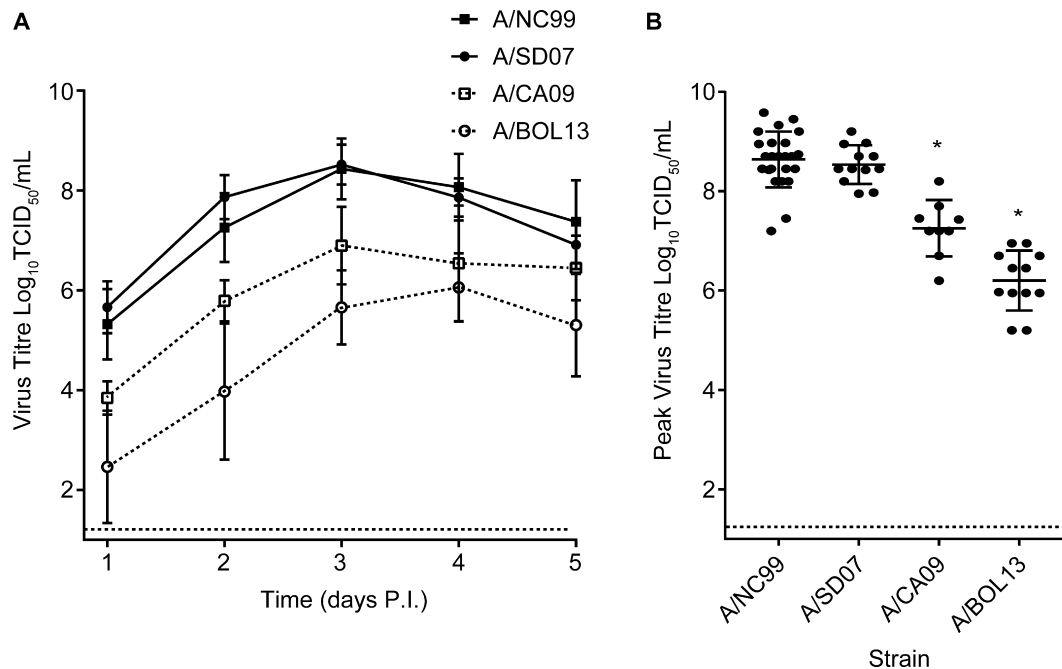


Fig. 4. Virus replication of H1N1 strains in human nasal epithelial cell (hNEC) cultures. hNEC cultures were infected with A/NC99, A/SD07, A/CA09 or A/BOL13 at a multiplicity of infection (MOI) of 0.1 for 5 days at 33 °C with apical wash collection every 24 h. Viral titres were determined by 50% tissue culture infectious dose (TCID₅₀) assay in Madin-Darby Canine Kidney (MDCK) cells. (A) Mean virus titres for each time point were generated from at least three independent experiments for each strain, each containing three separate transwells, and (B) peak viral titres were plotted. Error bars show ± 1 standard deviation and the limit of detection of the assay is shown by the dotted line. Differences between the peak viral titre of each strain were assessed by Tukey analysis (* $p < 0.05$ was considered as significant). A/BOL13 and A/CA09 had significantly lower peak titres than A/NC99 and A/SD07. P.I., post-infection.

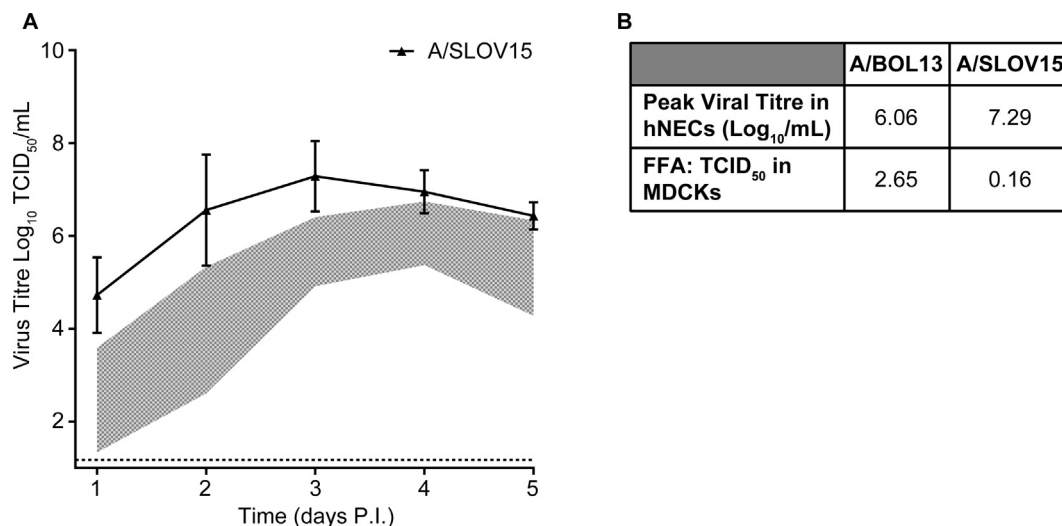


Fig. 5. Infectivity and virus replication of H1N1 strain A/SLOV15. (A) Human nasal epithelial cell (hNEC) cultures were infected with A/SLOV15 at a multiplicity of infection (MOI) of 0.1 for 5 days at 33 °C with apical wash collection every 24 h. Viral titres were determined by 50% tissue culture infectious dose (TCID₅₀) assay in Madin-Darby Canine Kidney (MDCK) cells. Mean virus titres for A/SLOV15 at each time point were generated from at least three independent experiments, each consisting of three separate transwell inserts. Error bars show ± 1 standard deviation for A/SLOV15 and the limit of detection of the assay is shown by the dotted line. The shaded area indicates one ± 1 standard deviation for A/BOL13 measurements. (B) Summary of peak viral titre in hNEC cultures and fluorescent focus assay (FFA):TCID₅₀ ratio in MDCK cells for A/BOL13 and A/SLOV15. P.I., post-infection.

the full extent of differences between pre-2009 and A/H1N1pdm09 viruses. Concurrently, this highlights the importance of using TCID₅₀ as an additional tool alongside FFA in the strain characterisation process, to ensure no deficiencies in multi-cycle infectivity are present. Understanding this relationship prior to commercial manufacture will enable the continued, robust use of FFA for vaccine release testing.

These data also indicate that LAIV replication in MDCK cells may not always be an accurate surrogate for replicative fitness in

the human nasal airway. Historically, both wt and LAIV viruses have been widely studied in non-human cell lines and in human cell lines of the lower respiratory tract [28–30]. Previous work by Ilyushina et al used cell models to assess the infectivity and attenuation of LAIVs for human use. The authors concluded that MDCK cells were less predictive than adenoid or human bronchial epithelial cells for displaying viruses with restricted replication [31]. This study highlights the importance of using a relevant cell model system when characterising LAIV replicative fitness. Primary hNEC

cultures most closely represent the environment of the nasal airway epithelia [22,32,33] and our experiments were able to clearly differentiate clinically effective pre-2009 A/H1N1 LAIV strains from the less fit A/H1N1pdm09 LAIV strains that had demonstrated reduced VE.

Another potential outcome of varying A/H1N1pdm09 fitness is in changes to the innate immune response to LAIV infection. Fischer et al showed that while LAIV strains generate less infectious virus in hNECs than their antigenically matched wt strains, they induce stronger innate immune activation [34]. An implication for our work is that a less fit strain, such as A/BOL13, might actually generate stronger innate immune responses than a more fit strain. However, it is worth noting that the Fischer comparison of LAIV and antigenically matched wt viruses may not be directly relevant to our studies. The majority of a LAIV gene constellation, including the interferon antagonist NS1 gene [35], is distinct from its antigenically matched wt virus. Thus, differences in interferon activation between LAIV and matched wt may not be surprising. Direct comparison of LAIV strains with varying replication but consistent internal gene constellations would require additional investigation, which could prove important for understanding the impact of LAIV fitness on immune activation. The underlying mechanism for the low replicative fitness of A/CA09 and A/BOL13 is currently under investigation. One explanation for this impaired replication could be the presence of non-infectious or partially infectious virus particles as suggested by the increased FFA:TCID₅₀ ratios. Recently, Gould et al suggested that LAIV contained defective interfering viral RNA sequences in both its influenza A and influenza B components and speculated that this could result in decreased immunising efficacy [36]. However, data produced in our laboratory refute this hypothesis, as we have shown that LAIV strains with high clinical efficacy also produce significant quantities of defective particles [37]. Despite this, a non-defective interfering-based assembly defect in viruses with reduced VE is still to be explored. As pre-2009 H1N1 and A/H1N1pdm09 strains possess different HA and NA genes but the same internal genes from the MDV backbone, a possible alternative explanation is that a HA/NA imbalance could cause differences in replicative fitness. Thus, balanced HA–NA functions are necessary for efficient influenza replication [38]; an imbalance in HA receptor binding and NA activity could cause a high level of receptor binding but not release or, conversely, an increased level of budding and release of virus prior to correct packaging of replicative material. Further studies are required to examine these ideas in more detail.

It should be noted that the H3N2 A/PAN99 also showed signs of a reduced FFA:TCID₅₀ ratio. However, unlike the A/H1N1pdm09 strains, the difference in the two measurements was not statistically significant due to variability of TCID₅₀ data. The replicative fitness of A/PAN99 in hNECs remains to be determined. It is unlikely that the observed FFA:TCID₅₀ ratio alone negatively affected A/PAN99, as it demonstrated significant efficacy in clinical trials [39,40]. Significant H3N2 VE has also been seen against more recent, antigenically matched strains [8]. In seasons where H3N2 VE was low, decoupling the contributions of vaccine strain mismatch and LAIV infectivity and replicative fitness will likely prove challenging. Meanwhile, as LAIV B virus VE tends to be robust [7,8], the lack of low VE data makes assessment of the relationship between fitness and VE difficult. It is also unclear at this stage whether subtype-specific differences in fitness or FFA:TCID₅₀ ratio might exist between LAIV viruses. While outside the scope of this paper, it will be essential to better understand the significance of both FFA:TCID₅₀ ratio and replicative fitness in human cells to VE of both H3N2 and B viruses.

Finally, lower replicative fitness could potentially result in the A/H1N1pdm09 strains being outcompeted by the presence of the other, fitter, vaccine strains in the QLAIV formulation. Delayed

replication kinetics, as seen for A/H1N1pdm09 strains in A549 and MDCK cells, could render these viruses more susceptible to competition at early stages post-infection. This could potentially cause a reduction in the generation of an immune response for strains with lower replicative fitness, leading to reduced protection. Ongoing work in our laboratory will help to determine the extent to which competition or interference occurs in the context of poorly replicating vaccine viruses. However, when responses to A/H1N1pdm09 strains were compared in a recent paediatric shedding and immunogenicity study, no differences in response were seen between trivalent LAIV and QLAIV formulations [33]. This indicated that QLAIV-specific interference was not the cause of reduced H1N1 VE. In addition, we know that reduced H1N1 VE was observed for A/CA09 in the US market when formulated as a trivalent vaccine in the 2010–2011 season and as a quadrivalent vaccine in the 2013–2014 influenza season [7], which argues against the transition to the quadrivalent formulation as the primary root cause of the reduced H1N1 effectiveness.

Following our investigation into the reduced effectiveness seen with the A/CA09 and A/BOL13 strains, we have added several new assays into the strain selection process. Vaccine viruses have traditionally been selected based on replication in eggs and MDCK cells. Additional characterisation assays such as primary hNEC infectivity and replication as measured by multiple assays such as FFA, TCID₅₀ and quantitative polymerase chain reaction have been utilised to improve vaccine candidate selection. Using these assays, we identified a new H1N1 strain, A/SLOV15, which had significantly improved multi-cycle infectivity and replicative fitness in hNECs, when compared with A/BOL13. A/SLOV15 was therefore selected for inclusion in the commercial 2017–2018 QLAIV formulation. Subsequently, a paediatric clinical shedding and immunogenicity study of this formulation was executed and showed significantly higher seroconversion rates for A/SLOV15 than for A/BOL13 [41]. Moreover, seroconversion rates for A/SLOV15 paralleled those of the highly efficacious pre-2009 H1N1 strain, A/NC99. This suggested that increased replicative fitness *in vitro* resulted in improved immunogenicity *in vivo*. These data contributed to the Advisory Committee on Immunization Practices voting in favour of renewing the recommendation for FluMist Quadrivalent® use in the US for the 2018–2019 season [42]. Finally, recent data from the 2017–2018 influenza season, using LAIV containing the new strain A/SLOV15, showed H1N1 VE at 90.3% for vaccine recipients aged 2–17 years in the UK (95% CI: 16.4–98.9) [43]. These data support the conclusion that the improvements in LAIV strain selection have delivered improved replicative fitness [15], resulting in the identification of an effective A/H1N1pdm09 LAIV strain.

In conclusion, our results demonstrate that A/H1N1pdm09 strains, A/CA09 and A/BOL13, have decreased replicative fitness in hNECs compared with pre-2009 A/H1N1 strains, demonstrating a plausible root cause for their reduced VE. This decreased replicative fitness has subsequently been corroborated by clinical data. hNEC cultures are now used to select novel H1N1 strains for the annual seasonal vaccine formulation. This tool, among others, provides a more accurate correlation to clinical VE, increasing confidence in the selection of vaccine strains to provide influenza illness protection for children and adults alike.

CRediT authorship contribution statement

Amy Hawksworth: Methodology, Formal Analysis, Writing – Original Draft, Visualization, Project Administration, Writing – Review & Editing. **Robert Lockhart:** Methodology, Formal Analysis, Visualization, Writing – Review & Editing. **Jonathan Crowe:** Methodology, Software, Validation, Formal Analysis, Data Curation,

Visualization, Writing - Review & Editing. **Ruben Maeso**: Methodology, Formal Analysis, Visualization, Writing - Review & Editing. **Lydia Ritter**: Investigation, Writing - review & editing. **Oliver Dibben**: Conceptualization, Formal Analysis, Writing - Original Draft, Writing - Review & Editing. **Helen Bright**: Conceptualization, Funding Acquisition, Writing - Review & Editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: At the time of data generation for this publication, Amy Hawksworth, Robert Lockhart, Jonathan Crowe, Ruben Maeso, Lydia Ritter, Oliver Dibben and Helen Bright were employees of MedImmune UK Ltd (wholly owned subsidiary of AstraZeneca plc). MedImmune UK Ltd are the manufacturers of Fluenz Tetra®/FluMist Quadrivalent® intranasal influenza live virus vaccine.

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